

BASAL NUCLEI (GANGLIA)



Learning Objectives

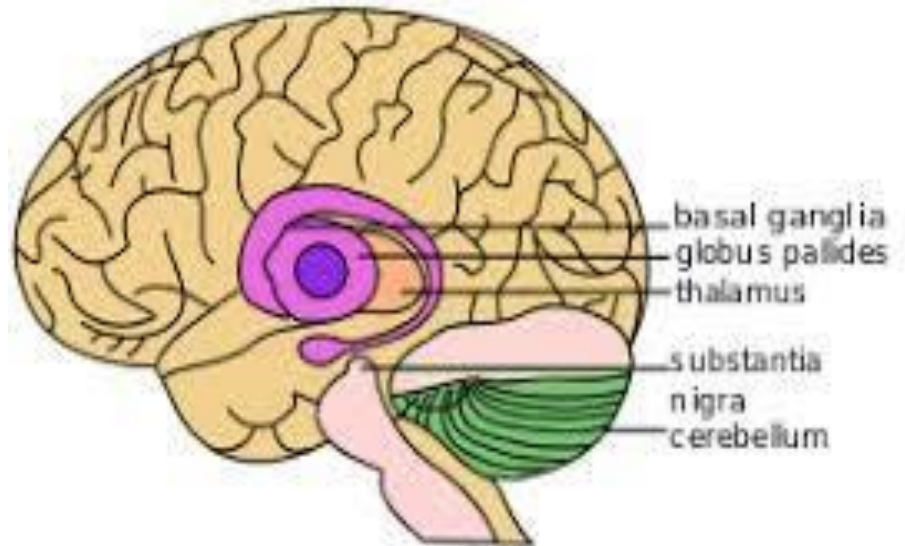
At the end of the lecture the students should be able to:

- Define basal nuclei.
- Describe the anatomy & functions of basal ganglia.
- Enlist the components of basal nuclei.
- Define parts of caudate nucleus.
- Describe connections & important disorders of basal nuclei.

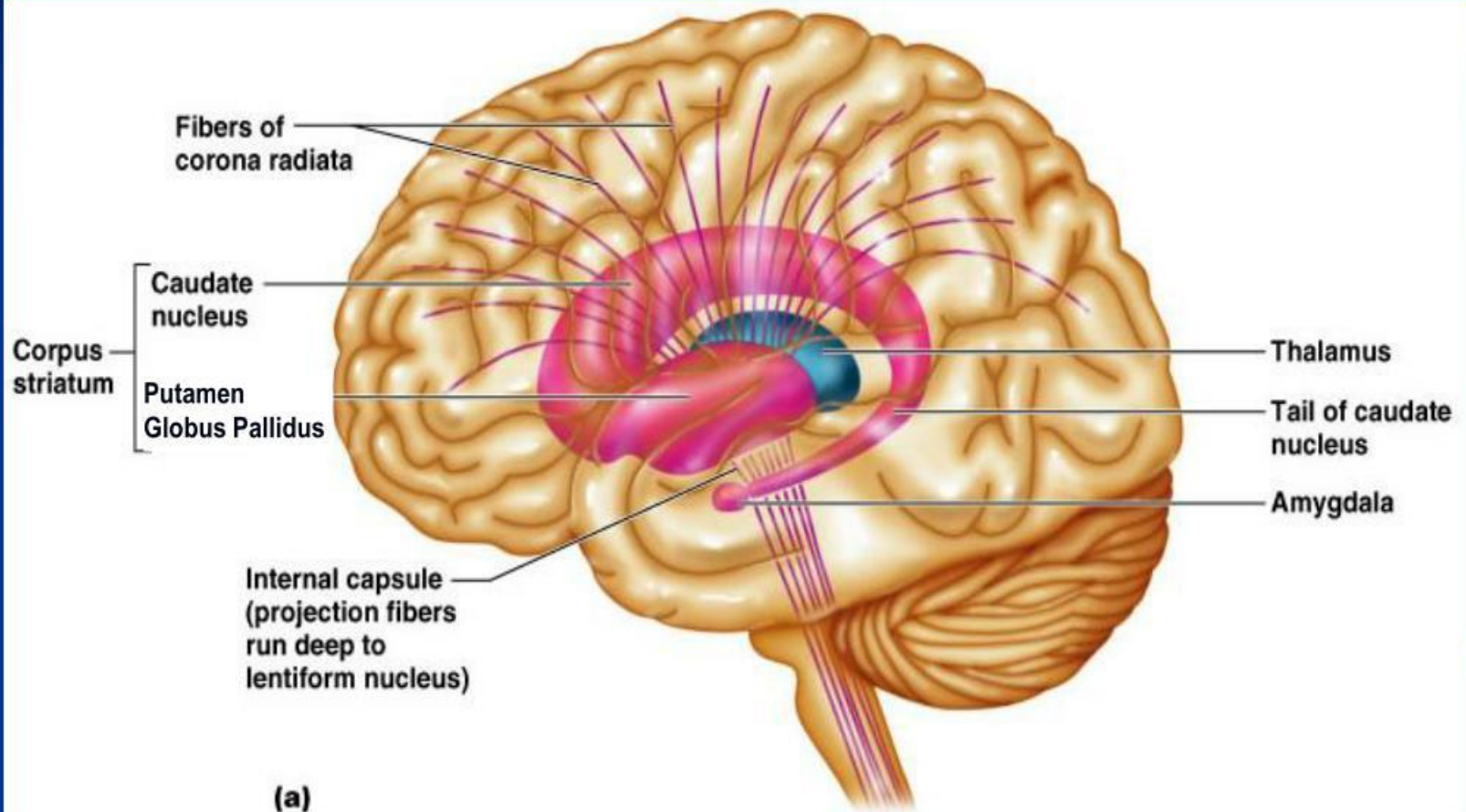
Introduction

- Basal nuclei refer to a group of nuclei located deep within cerebral hemispheres.
- They are part of extrapyramidal motor system, principally involved in the control of posture & voluntary movements mainly by inhibiting motor functions.
- They include corpus striatum, amygdaloid nucleus & claustrum.
- Has important connections with other parts of brain such as thalamus, sub-thalamic nuclei, red nucleus & substantia nigra.

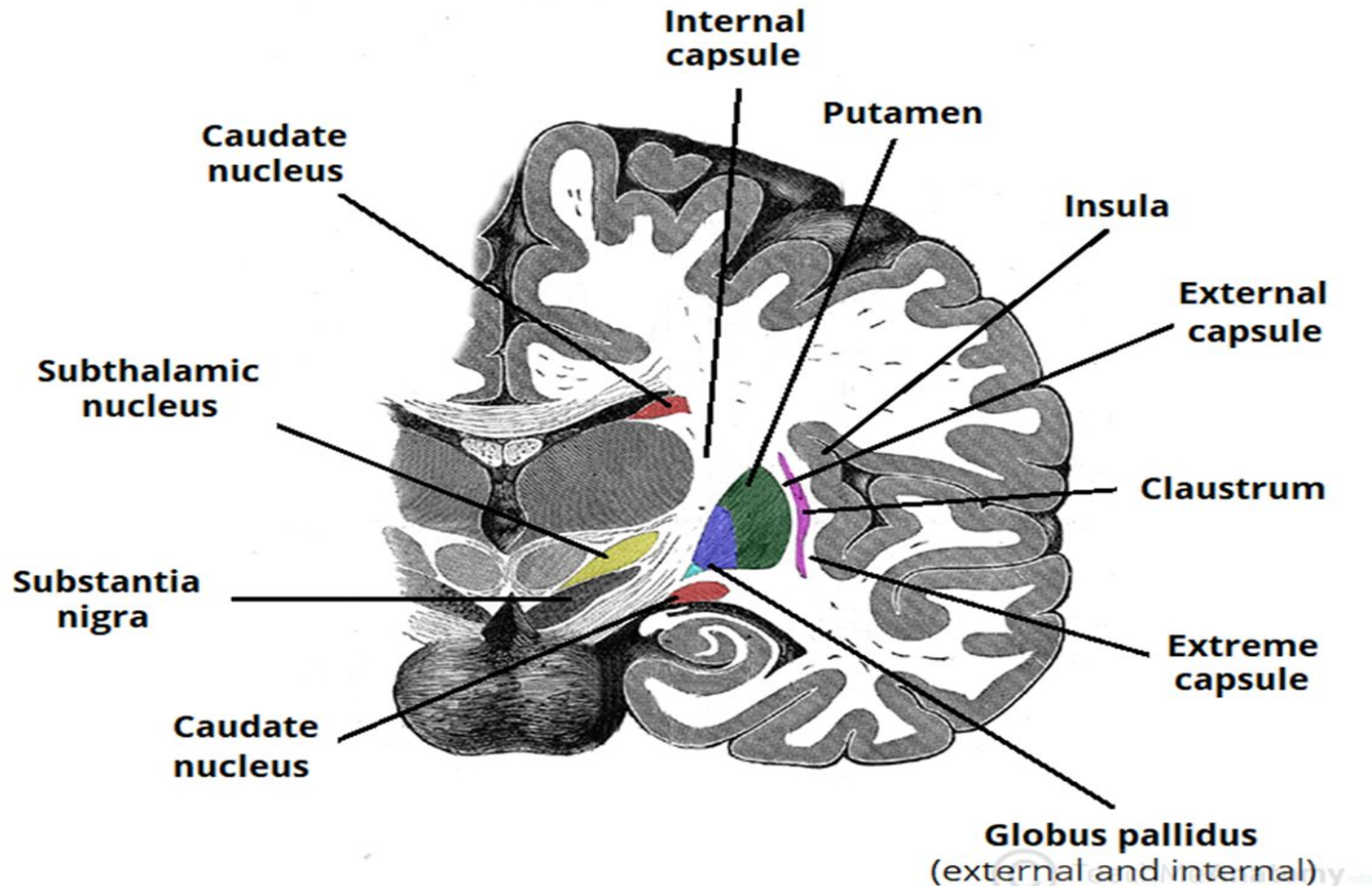
Basal Ganglia and Related Structures of the Brain



Basal Ganglia



Basal nuclei/ganglia viewed via coronal section of brain

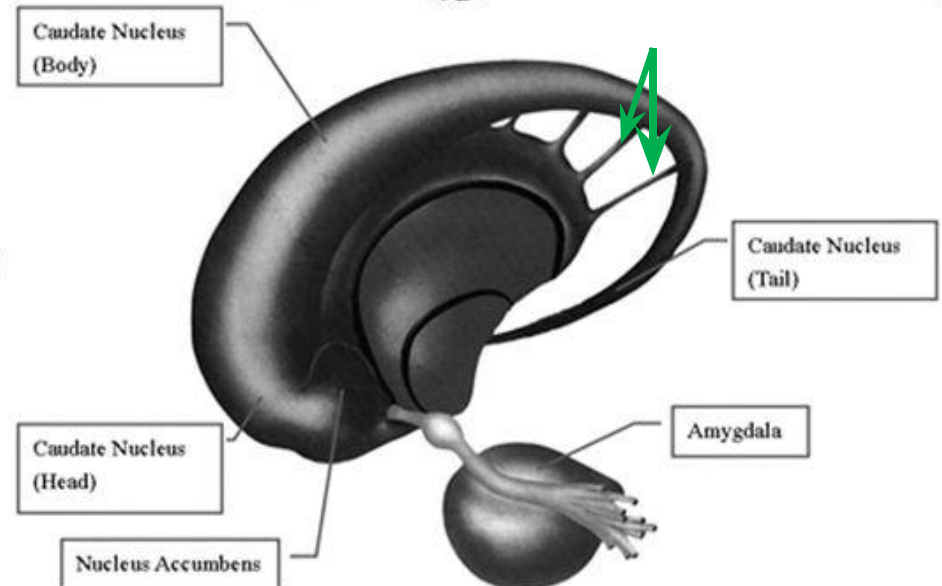
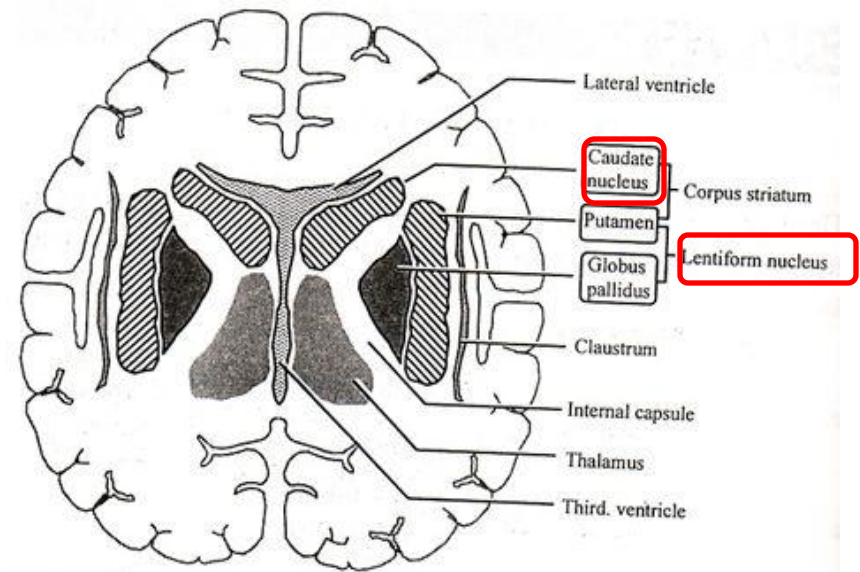


Terminologies

Neurological Structure	Basal Nuclei
Corpus striatum	Caudate nucleus + lentiform nucleus
Caudate nucleus	Caudate nucleus
Lentiform nucleus	Globus pallidus + putamen
Amygdala	Amygdaloid nucleus
Clastrum	Clastrum
Neostriatum	Caudate nucleus + putamen
Paleostriatum	Globus pallidus

Corpus Striatum

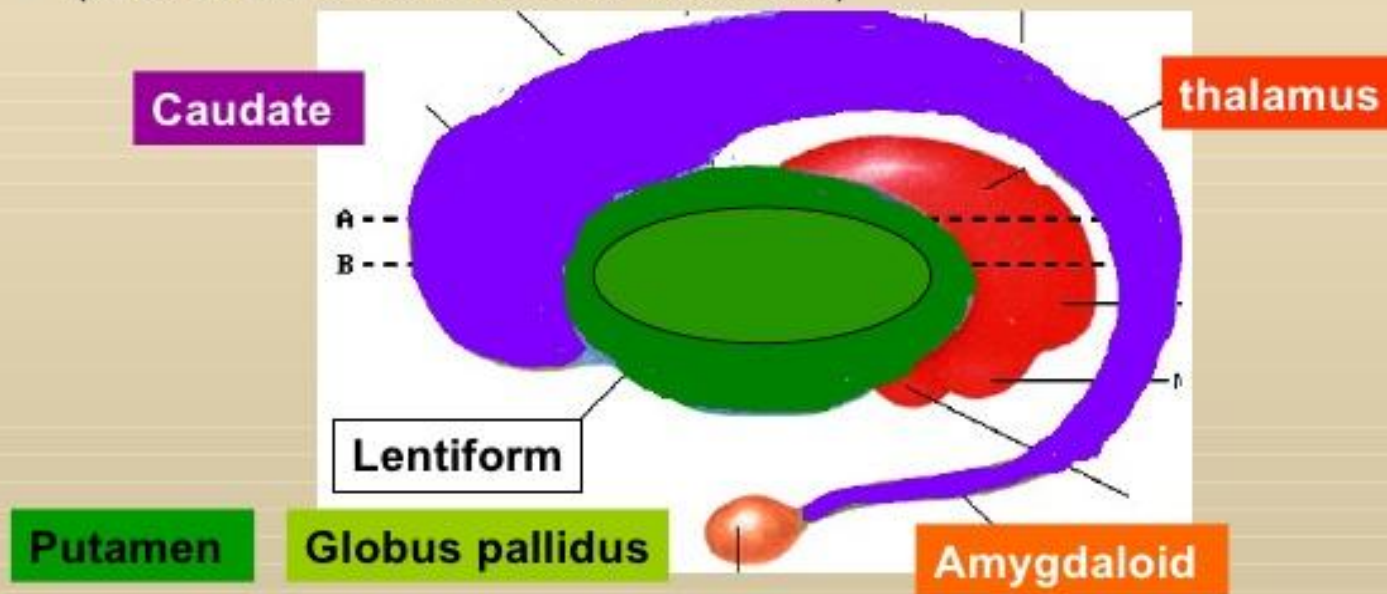
- Lies lateral to the thalamus.
- It is divided by **internal capsule** into **caudate nucleus** and **lentiform nucleus**.
- Bands of **grey matter** pass from **lentiform nucleus** across the internal capsule to the **caudate nucleus**, giving the striated appearance hence, the name **corpus striatum**.
- The most ventral part of the corpus striatum is called the **nucleus accumbens**, which has connections with the limbic system.



Caudate nucleus

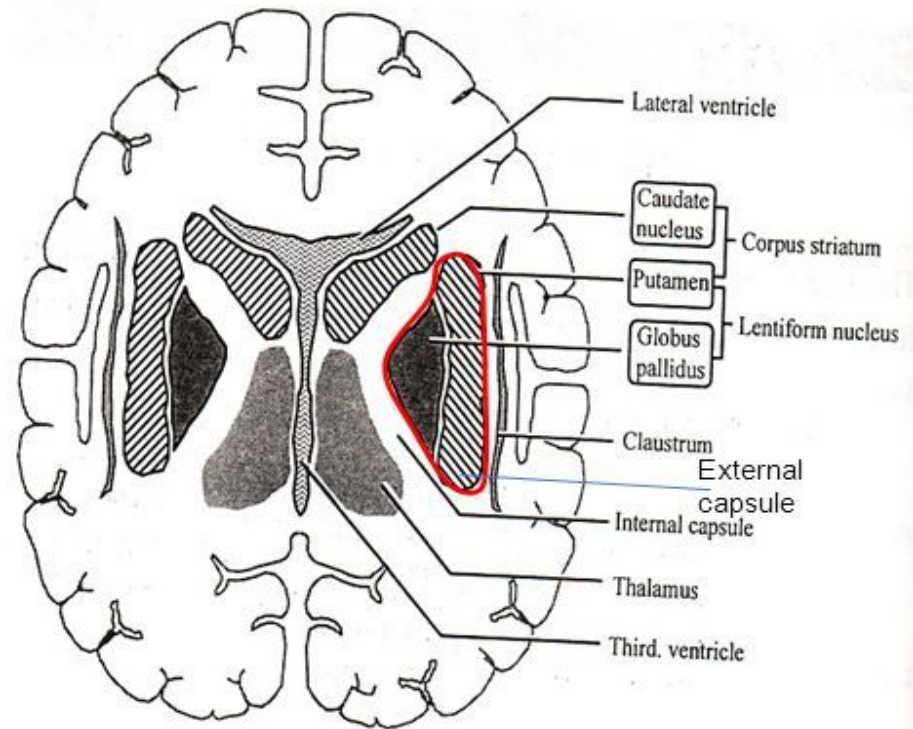
- C-shaped
- Head, body, tail
- Large head, tapering curved tail
- Head-frontal lobe
- Tail-occipital lobe
- End of tail-temporal lobe
 - -terminates in amygdaloid nucleus
 - (roof of inf horn of lateral ventricle)

Caudate nucleus



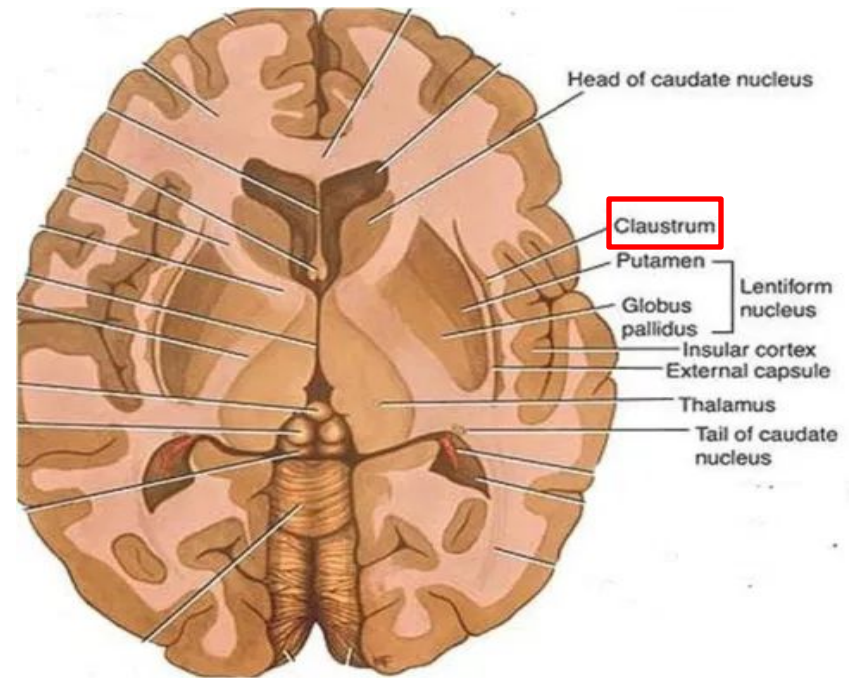
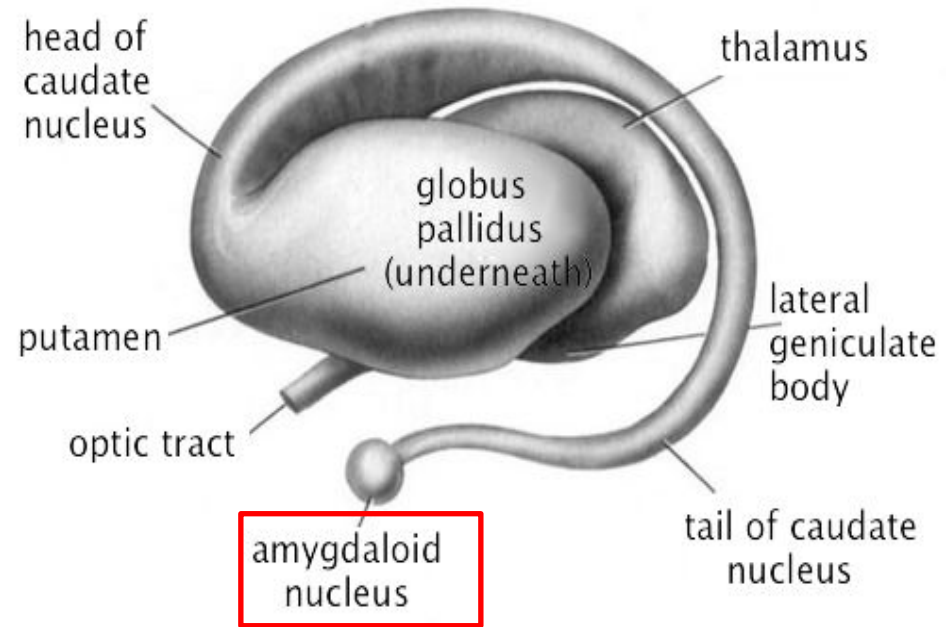
Lentiform Nucleus

- A three sided, wedge-shaped mass of grey matter, with a convex outer surface and an apex which lies against the **genu** of the internal capsule
- Divided into a:
 - Larger darker lateral portion called **Putamen** &
 - Smaller, lighter medial portion called **Globus pallidus**



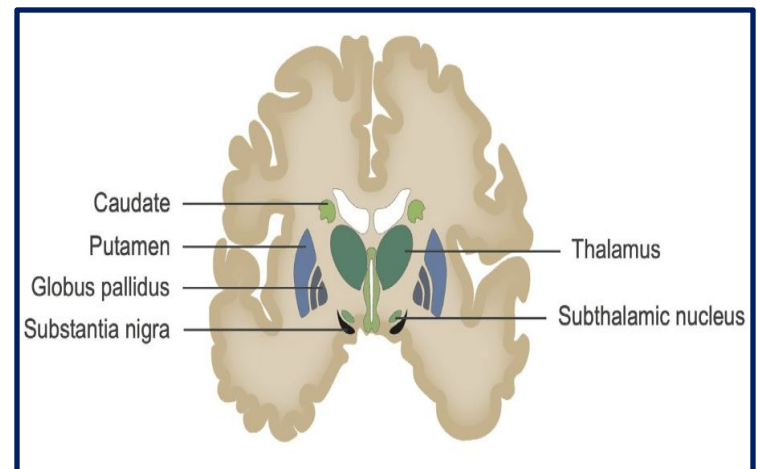
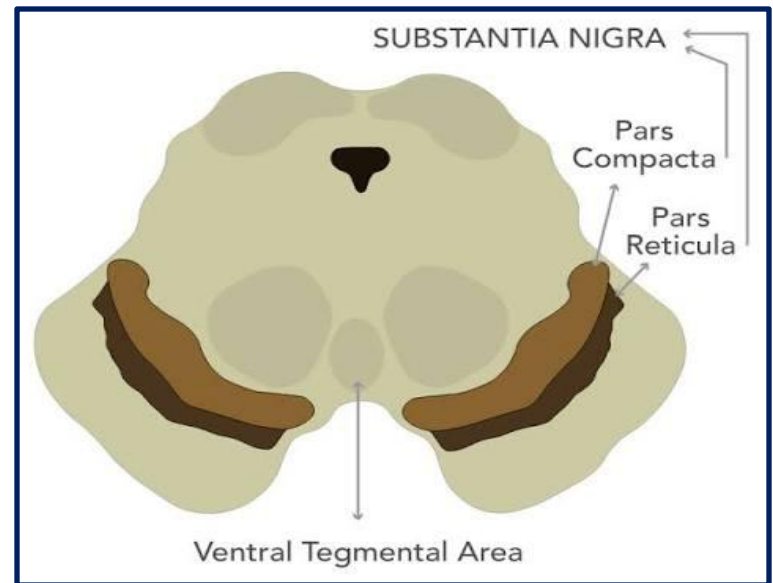
Amygdaloid nucleus: is situated in the temporal lobe.

- It is considered to be part of the limbic system.
- It can influence the body's response to environmental changes e.g. in the sense of fear, it can alter the heart rate, blood pressure, skin color and respiratory rate etc.
- **Clastrum:** is a thin sheet of gray matter which is separated from the lateral surface of lentiform nucleus by external capsule.



Substantia Nigra & Subthalamic Nuclei:

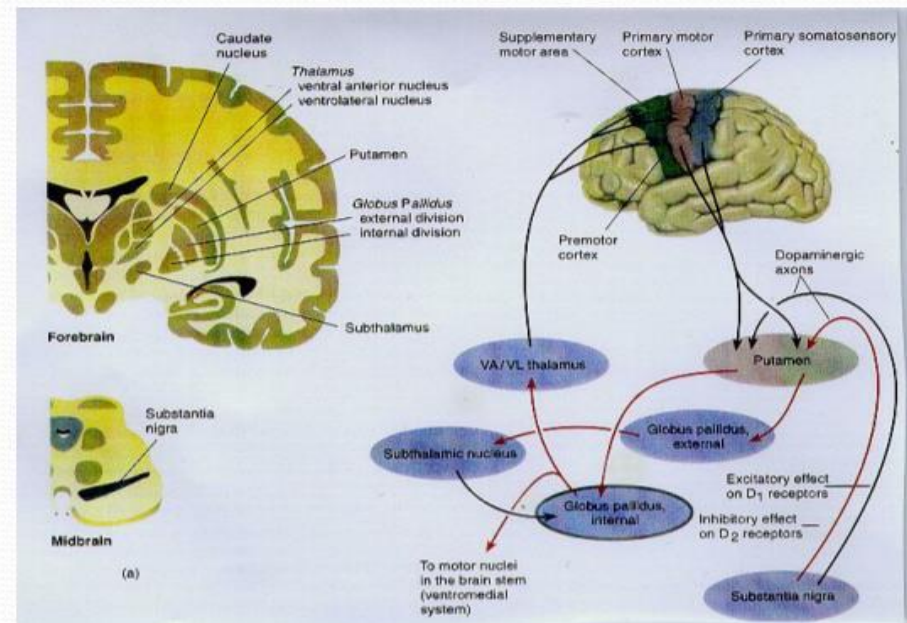
- The substantia nigra & subthalamic nuclei are functionally closely related to the basal nuclei.
- The substantia nigra pars compacta (SNc) contains densely packed, dark-pigmented dopaminergic neurons that release dopamine to facilitate movement.
- Whereas, the pars reticulata (SNr) contains dispersed GABAergic neurons that act as an output center relaying inhibitory signals from the basal ganglia to other brain structures.
- The substantia nigra has connections to the corpus striatum.
- It consists of dopaminergic and inhibitory neurons.
- The subthalamic nuclei has connections to the globus pallidus & substantia nigra.
- They have glutaminergic and excitatory neurons.



Connections of the Corpus Striatum & the Globus Pallidus

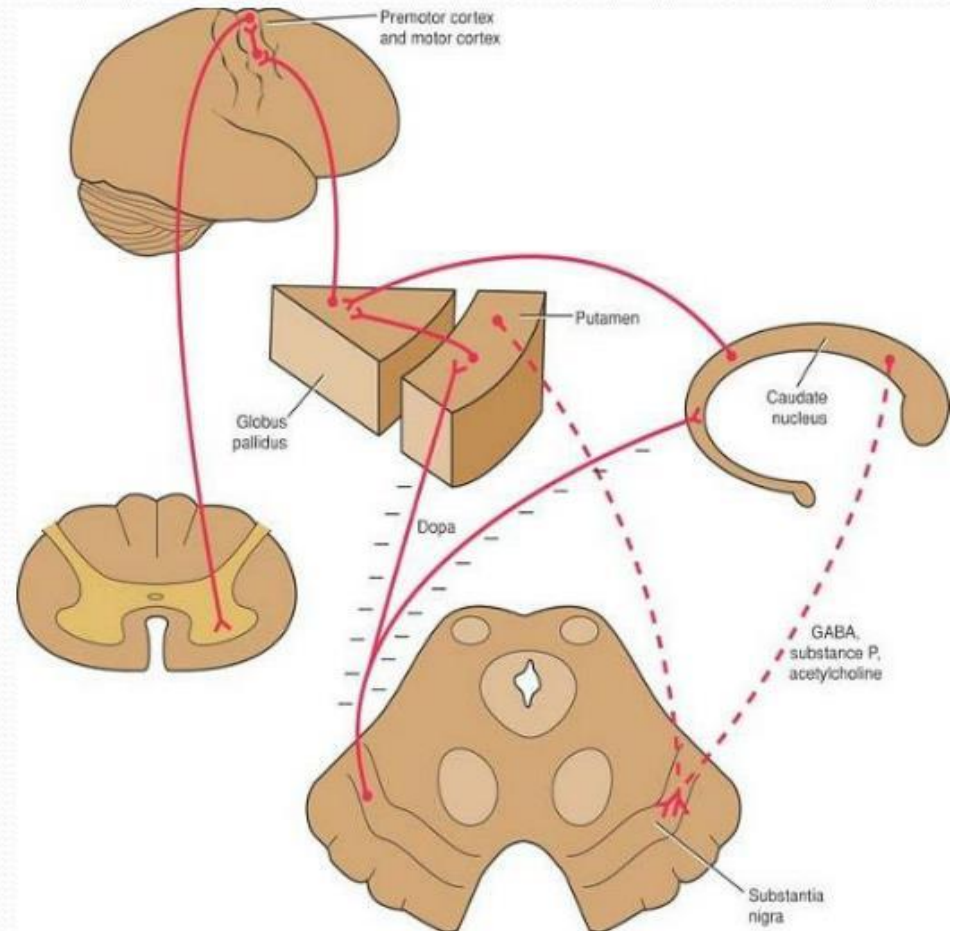
Connections of the Corpus Striatum:

- **Connections:**
- **Caudate nucleus and putamen** are main sites for receiving **input** to the basal nuclei.
- The **globus pallidus** is major site of **output** from basal nuclei.
- **Connections of the Corpus Striatum:**
- **Afferent fibers:**
- **Corticostriate.**
- **Thalamostriate.**
- **Nigrostriate.**
- **Brainstem.**



Connections of the Globus Pallidus:

- **Efferent fibers:**
- **Striatopallidal.**
- **Striatonigral.**
- **Connections of the globus pallidus:**
- **Afferent:**
- **Striatopallidal.**
- **Efferent :**
- **Pallidofugal Fibers** which are:
- **Ansa lenticularis** to thalamic nuclei.
- **Fasciculus lenticularis** to subthalamus.
- **Pallidosubthalamic** to subthalamic nuclei.
- **Pallidotegmental** to tegmentum of the midbrain.

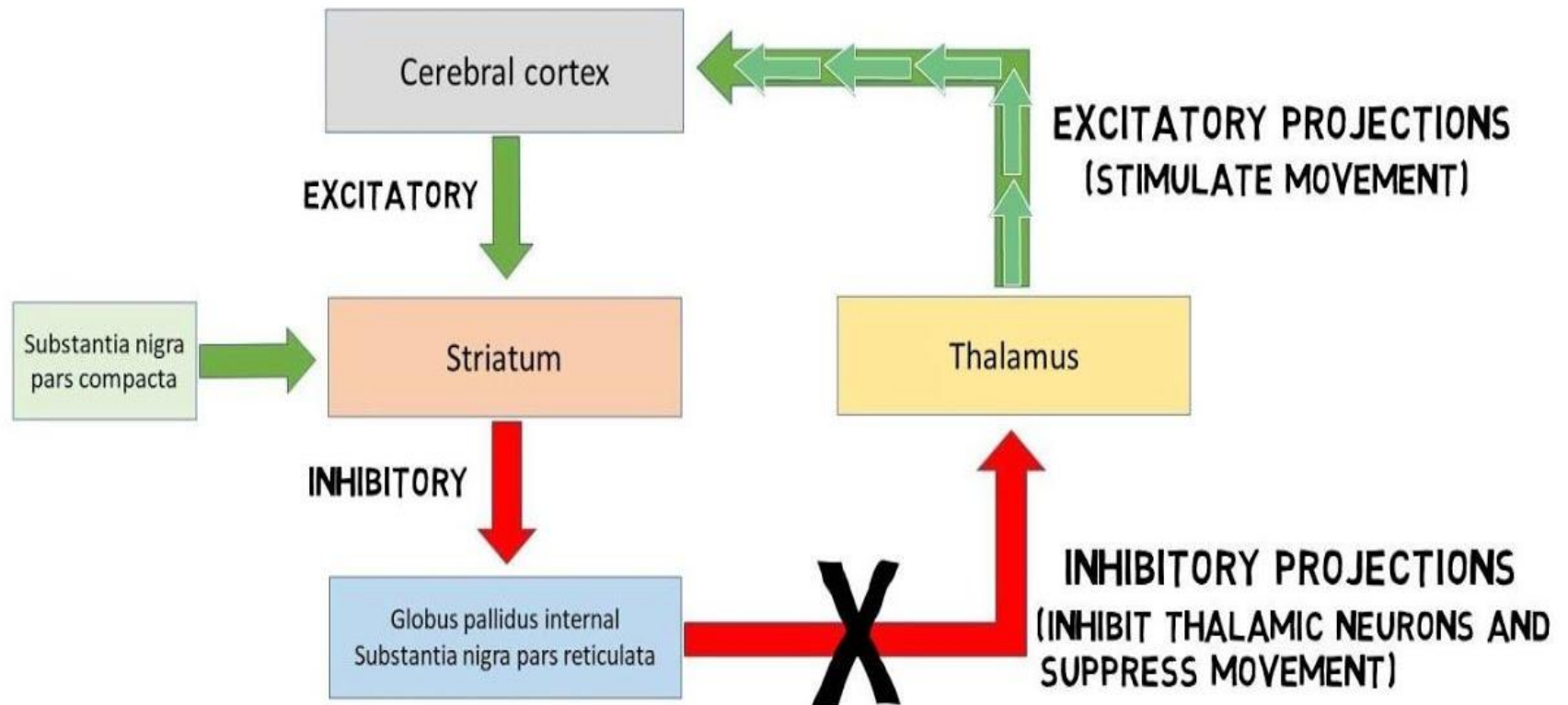


Direct Pathway of Basal Nuclei:

- **Definition:** The direct pathway of the basal ganglia is a neural circuit that facilitates and initiates voluntary movement by disinhibiting the thalamus.
- The direct pathway flows in a specific order through the brain:
- ✓ **Cerebral Cortex:** Sends an excitatory signal using glutamate to the striatum.
- ✓ **Striatum (Caudate and Putamen):** Sends an inhibitory signal using GABA to the internal globus pallidus and substantia nigra.
- ✓ **Globus Pallidus Internus (GPi) and Substantia Nigra pars reticulata (SNr):** Normally inhibit the thalamus, but are themselves inhibited (turned off) by the striatum
- ✓ **Thalamus:** Becomes disinhibited (released from inhibition), allowing it to send excitatory signals back to the motor cortex.
- ✓ **Motor Cortex:** Excites spinal motor neurons to execute the movement.
- Net Effect: Excites motor cortex & thus promotes movement.

DIRECT PATHWAY

(SIMPLIFIED MODEL)



Indirect Pathway of Basal Nuclei:

- **Definition:** The indirect pathway of the basal ganglia is a neural circuit that inhibits unwanted muscle movements and suppresses competing motor programs.
- The indirect pathway steps are as follows:
 - ✓ **Cortex to Striatum:** The motor cortex sends excitatory signals (using glutamate) to the striatum.
 - ✓ **Striatum to GPe:** The striatum sends inhibitory signals (using GABA) to the external segment of the globus pallidus (GPe).
 - ✓ **GPe to Subthalamic Nucleus:** Inhibiting the GPe means it stops releasing its normal inhibitory signals onto the subthalamic nucleus (STN). This removal of inhibition is called disinhibition.
 - ✓ **Subthalamic Nucleus to GPi/SNr:** Freed from GPe control, the subthalamic nucleus becomes active and sends excitatory signals (glutamate) to the internal segment of the globus pallidus (GPi) and the substantia nigra pars reticulata (SNr).
 - ✓ **GPi/SNr to Thalamus:** The GPi and SNr then fire strong inhibitory signals (GABA) to the thalamus.
 - ✓ **Thalamus to Cortex:** Because the thalamus is inhibited, it cannot send excitatory signals up to the motor cortex.
 - **Net Result:** Movement is suppressed.
 - **Function:** Stops unwanted movements.

(SIMPLIFIED MODEL)

```
graph TD
    CC[Cerebral cortex] -- red --> ST[Striatum]
    ST -- red --> GPe[Globus pallidus external]
    GPe -- red --> GPi[GPi]
    GPi -- red --> SNr[SNr]
    SNr -- red --> TH[Thalamus]
    TH -- red --> CC
    
    CC -- green --> ST
    ST -- green --> GPe
    GPe -- green --> STN[Subthalamic nucleus]
    STN -- green --> GPi
    GPi -- green --> TH
    TH -- green --> CC
```

Functions of basal ganglia

- Function in association with the corticospinal system to **control complex patterns of motor activity**(cutting paper with scissors, hammering nails, shooting a basketball through a hoop, passing a football, throwing a baseball etc.
- **Cognitive control of motor activity** e.g. if a person comes across a lion he will subconsciously plan 1) turning away from the lion, (2) beginning to run, and (3) even attempting to climb a tree
- To Change the Timing and to Scale the Intensity of Movements

Disorders of Basal ganglia

Hypokinetic disorders

Parkinsonism

Hyperkinetic disorders

1. **A**thetosis
2. **B**allismus
3. **C**horea
4. **T**orsion spasm

Effects of Basal Ganglia Lesions		
Facilitation of motor movements. ♥		
INVOLUNTARY MOVEMENTS	LESION SITE	FEATURES
 Chorea	 Caudate nucleus	★ Rapid, involuntary "dancing" movements
 Athetosis	 Globus pallidus & putamen	★ Continuous, <u>slow</u> writhing movements
 Wing-beating tremors	 Hepatolenticular degeneration	★ Seen in <u>Wilson's disease</u> (↑ copper deposition)
 Hemiballismus	 Subthalamic nucleus (STN)	★ Sudden, <u>violent</u> flailing movements of one large joint
 Pill-rolling tremor (resting tremor)	 Substantia nigra pars compacta (SNpc)	★ Seen in <u>Parkinson's disease</u> (↓ dopamine)

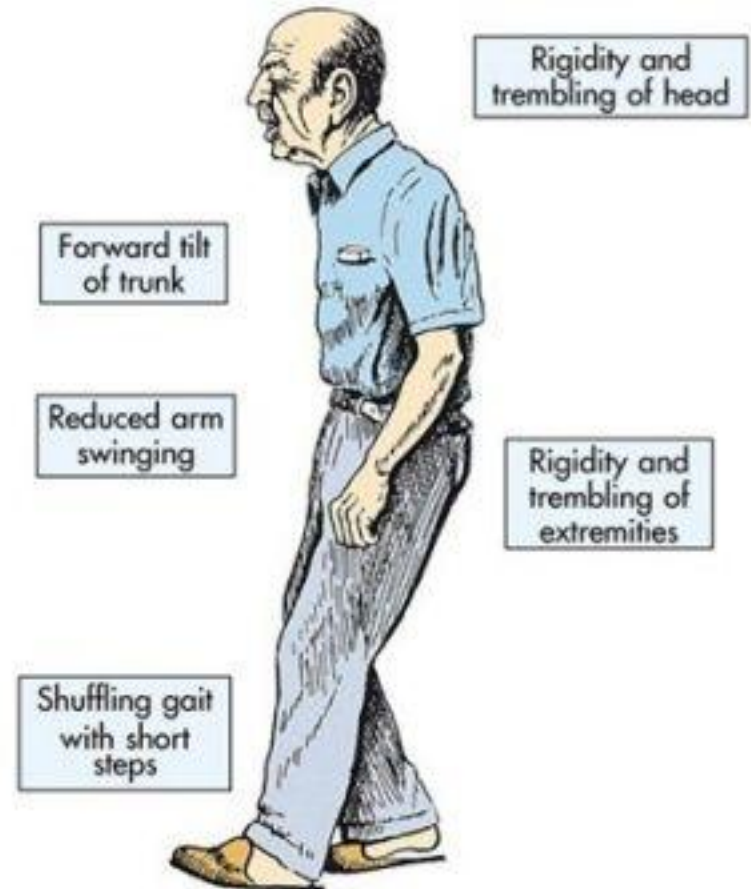
@mood_rx ♥

Disorders of the basal ganglia

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Parkinson's Disease

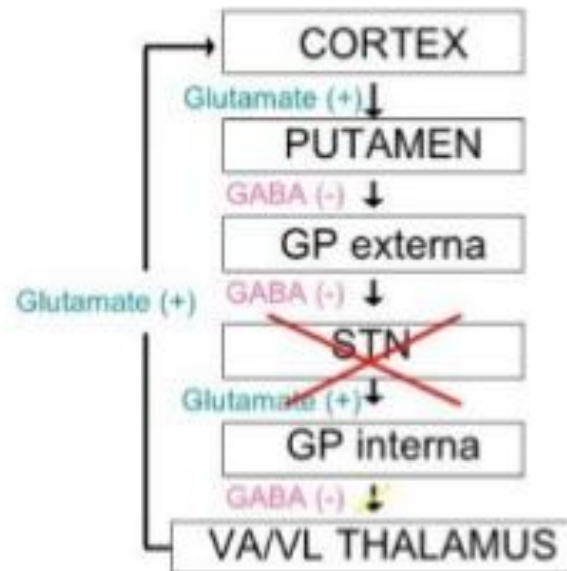
- Characterized by resting tremor, slowed/absent movement (**hypokinesia**), rigidity of the extremities and neck, & reduced facial expressiveness
- Caused by the loss of the dopaminergic neurons in the substantia nigra pars compacta
- Typically treated with L-Dopa



WebCT readings: Circuits within the basal ganglia system

Disorders of Basal ganglia

- **Chorea (dancing disease)**
 - the patient exhibits involuntary, quick, jerky, irregular movements that are nonrepetitive.
 - Swift grimaces and sudden movements of the head or limbs are good examples.
- **Hemiballismus**
 - form of involuntary movement to one side of the body.
 - It usually involves the proximal extremity musculature, and the limb suddenly flies about out of control in all directions.
 - The lesion, which is usually a small stroke, occurs in the opposite subthalamic nucleus or its connections; it is in the subthalamic nucleus that smooth movements of different parts of the body are integrated.



References:

- **CLINICAL
NEUROANATOMY**
Richard S. Snell
8th edition
Chapter 10
Pages # 317 - 325